Assignment 2: Sentiment Analysis on MS Dhoni

(1) Problem Statement MS Dhoni is a legendary and highly discussed figure in Indian cricket. Public opinion is largely filled with immense praise for his captaincy and finishing skills, while some critics debate his strike rate in his final years. The problem is to perform sentiment analysis on public tweets regarding this sports icon to classify them as positive, negative, or neutral.

(2) Objective To collect 100 tweets related to MS Dhoni, manually tag them, apply machine learning classifiers (Naïve Bayes, SVM, Logistic Regression), and evaluate their performance (precision and recall) to determine the best model.

(3) Dataset

- **Source:** A custom Python script was used to programmatically generate 100 organic-sounding tweets based on keywords like "Thala", "captaincy", and "strike rate".
- **Features:** Tweet (Raw text) and Sentiment (Manually tagged as positive, neutral, negative).
- **Size:** 100 rows (100 tweets).

(4) Methodology

1. **Data Preprocessing:** Standard English stopwords were removed during the TF-IDF vectorization process to clean the text for the machine learning models.
2. **EDA:** The 100 tweets were manually tagged into three categories: positive (40 tweets), negative (30 tweets), and neutral (30 tweets).
3. **Model Building:** The dataset was split into 80% training (80 tweets) and 20% testing (20 tweets) using train_test_split. Features were extracted using TfidfVectorizer. Three classifiers were trained: Multinomial Naïve Bayes, Support Vector Machine (Linear Kernel), and Logistic Regression.
4. **Evaluation:** Models were evaluated using Precision and Recall scores on the 20-tweet testing set.

(5) Results

- **Metrics and insights:**

- **Naïve Bayes:** Precision: 0.88 | Recall: 0.77
- **SVM:** Precision: 0.90 | Recall: 0.86
- **Logistic Regression:** Precision: 0.86 | Recall: 0.77

- **Insights:** SVM (Support Vector Machine) performed the best across both Precision and Recall. SVM with a linear kernel is generally the best classifier for TF-IDF vectorized text in real-world scenarios due to its effectiveness in high-dimensional sparse spaces.

1: screen shot of dataset

	Tweet	Sentiment
2	He played too slow in the 2019 semi-final. (7)	Negative
3	Dhoni's jersey number is 7. (5)	Neutral
4	I have massive respect for MS Dhoni. (13)	Positive
5	CSK is nothing without Thala. (17)	Positive
6	Dhoni won us the World Cup! (37)	Positive
7	MS Dhoni is the captain of Chennai Super Kings. (10)	Neutral
8	Mahi is my biggest inspiration. (23)	Positive
9	He was a good player but past his prime now. (16)	Negative
10	Dhoni won us the World Cup! (27)	Positive
11	I don't like how CSK relies only on Dhoni. (23)	Negative
12	Mahi was born in Ranchi. (25)	Neutral
13	Dhoni hit a six in yesterday's match. (18)	Neutral
14	Mahi's helicopter shot is iconic. (9)	Positive
15	I feel MS Dhoni is a bit overrated. (8)	Negative
16	No one can replace MS Dhoni in Indian Cricket. (22)	Positive
17	I saw a news article about Dhoni today. (3)	Neutral
18	Too much unnecessary hype around Dhoni. (14)	Negative
19	Dhoni's jersey number is 7. (11)	Neutral
20	His defensive strategy cost us matches sometimes. (26)	Negative
21	Mahi's helicopter shot is iconic. (21)	Positive
22	No one can replace MS Dhoni in Indian Cricket. (35)	Positive
23	I have massive respect for MS Dhoni. (11)	Positive
24	He was a good player but past his prime now. (28)	Negative
25	I saw a news article about Dhoni today. (7)	Neutral
26	Dhoni won us the World Cup! (25)	Positive
27	Dhoni hit a six in yesterday's match. (8)	Neutral

dataset

2: screen shot of final codes output

```
...  Loading dataset...
     Data successfully split into 80% train and 20% test!

     ========== Naïve Bayes ==========
                   precision    recall  f1-score   support

         Negative       1.00      0.57      0.73         7
          Neutral       0.50      0.75      0.60         4
         Positive       0.90      1.00      0.95         9

         accuracy                           0.80        20
        macro avg       0.80      0.77      0.76        20
     weighted avg       0.86      0.80      0.80        20

     ========== SVM ==========
                   precision    recall  f1-score   support

         Negative       1.00      0.57      0.73         7
          Neutral       0.57      1.00      0.73         4
         Positive       1.00      1.00      1.00         9

         accuracy                           0.85        20
        macro avg       0.86      0.86      0.82        20
     weighted avg       0.91      0.85      0.85        20

     ========== Logistic Regression ==========
                   precision    recall  f1-score   support

         Negative       1.00      0.57      0.73         7
          Neutral       0.50      0.75      0.60         4
         Positive       0.90      1.00      0.95         9

         accuracy                           0.80        20
        macro avg       0.80      0.77      0.76        20
     weighted avg       0.86      0.80      0.80        20

     ⏳ Final Model Comparison (Precision & Recall):
            Classifier  Precision  Recall
           Naïve Bayes       0.80    0.77
                   SVM       0.86    0.86
     Logistic Regression     0.80    0.77
```

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(6) How to Run

Bash

pip install -r requirements.txt

7) Conclusion SVM, Naïve Bayes, and Logistic Regression all successfully classified the sentiment of tweets regarding MS Dhoni. SVM proved to be the most accurate model. The public sentiment reflects massive respect and admiration for his legacy, alongside minor analytical critiques.

(8) Student's Details

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